

Vi skal ”bare” isolere x i ligningen

$$a \cdot x^2 + b \cdot x + c = 0$$

Alle mellemregninger står her.

$$a \cdot x^2 + b \cdot x + c = 0$$

$$a \cdot x^2 + b \cdot x = -c$$

$$x^2 + \frac{b}{a} \cdot x = -\frac{c}{a}$$

$$\left(x + \frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 = -\frac{c}{a}$$

$$\left(x + \frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2$$

$$\left(x + \frac{b}{2a}\right)^2 = -\frac{c}{a} + \frac{b^2}{2^2 a^2}$$

$$\left(x + \frac{b}{2a}\right)^2 = -\frac{c}{a} + \frac{b^2}{4a^2}$$

$$\left(x + \frac{b}{2a}\right)^2 = -\frac{4 \cdot a \cdot c}{a \cdot 4a} + \frac{b^2}{4a^2}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4 \cdot a \cdot c}{4a^2}$$

$$x + \frac{b}{2a} = \pm \sqrt{\frac{b^2 - 4 \cdot a \cdot c}{4a^2}}$$

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4 \cdot a \cdot c}}{\sqrt{4a^2}}$$

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4 \cdot a \cdot c}}{2a}$$

$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4 \cdot a \cdot c}}{2a}$$